INTRODUCTION

Investment decisions are made under uncertainty about the future direction of the economy, issuers, companies, and prices. This learning module presents probability tools that address many real-world problems involving uncertainty and applies to a variety of investment management applications.

Lesson 1 introduces the calculation of the expected value, variance, and standard deviation for a random variable. These are essential quantitative concepts in investment management. Lesson 2 introduces probability trees that help in visualizing the conditional expectations and the total probabilities for expected value.

When making investment decisions, analysts often rely on perspectives, which may be influenced by subsequent observations. Lesson 3 introduces Bayes' formula, a rational method to adjust probabilities with the arrival of new information. This method has wide business and investment applications.

Learning Module Overview

- The expected value of a random variable is a probability-weighted average of the possible outcomes of the random variable. For a random variable X, the expected value of X is denoted $E(X)$.
- The variance of a random variable is the expected value (the probability-weighted average) of squared deviations from the random variable's expected value $E(X)$: $\sigma^2(X) = E\{[X - E(X)]^2\}$, where $\sigma^2(X)$ stands for the variance of X.
- Standard deviation is the positive square root of variance. Standard deviation measures dispersion (as does variance), but it is measured in the same units as the variable.
- A probability tree is a means of illustrating the results of two or more independent events.
- A probability of an event given (conditioned on) another event is a conditional probability. The probability of an event A given an event B is denoted $P(A \mid B)$, and $P(A \mid B) = P(AB)/P(B)$, $P(B) \neq 0$.
- According to the total probability rule, if $S1$, $S2$, …, Sn are mutually exclusive and exhaustive scenarios or events, then $P(A) = P(A \mid S1)P(S1) + P(A \mid S2)P(S2) + \ldots + P(A \mid Sn)P(Sn)$.
- Conditional expected value is $E(X \mid S) = P(X_1 \mid S)X_1 + P(X_2 \mid S)X_2 + \ldots + P(X_n \mid S)X_n$ and has an associated conditional variance and conditional standard deviation.
- Bayes' formula is a method used to update probabilities based on new information.
- Bayes' formula is expressed as follows: Updated probability of event given the new information = [(Probability of the new information given event)/(Unconditional probability of the new information)] × Prior probability of event.

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